

ϕ^3 in $d=6-\epsilon$

$$\mathcal{L} = -\frac{1}{2} Z_\phi \partial^\mu \phi \partial_\mu \phi - \frac{1}{2} Z_m m^2 \phi^2 + Z_g g \tilde{\mu}^{3/2} \phi^3 + Y \phi$$

$$= -\frac{1}{2} \partial^\mu \phi_0 \partial_\mu \phi_0 - \frac{1}{2} m_0^2 \phi_0^2 + \frac{1}{6} g_0 \phi_0^3 + Y_0 \phi_0$$

ϕ : Renormalized fields

$g, \tilde{\mu}$: Renormalized parameters

• In $\overline{MS}$ with $\mu^2 = 4\pi e^{\gamma_E} \tilde{\mu}^2$

ϕ_0 : Bare fields

• The relationship between them

$$\phi_0 = Z_\phi^{1/2} \phi$$

• Plugging this into the second term:

$$Z_m m^2 Z_\phi \phi_0^2 \Rightarrow m_0^2 = Z_m Z_\phi^{-1} m^2$$

$$m_0 = Z_m^{1/2} Z_\phi^{-1/2} m$$

• Third term

$$\frac{Z_g \tilde{\mu}^{3/2} g Z_\phi^{3/2} \phi_0^3}{Z_g \tilde{\mu}^{3/2} g Z_\phi^{3/2}} = g_0 \phi_0^3$$

$$g_0 = Z_g Z_\phi^{3/2} g \tilde{\mu}^{3/2}$$

• Fourth term:

$$Y \phi = Y_0 \phi_0 \Rightarrow Z_\phi^{1/2} Y = Y_0$$

Collecting these we have

$$\phi_0(x) = Z_\phi^{1/2} \phi(x)$$

$$m_0 = Z_\phi^{-1/2} Z_m^{1/2} m$$

$$g_0 = Z_g Z_\phi^{3/2} g \tilde{\mu}^{3/2}$$

$$Y_0 = Z_\phi^{-1/2} Y$$

• For $\Pi(k^2)$: $A \sim \frac{1}{\epsilon} + \mathcal{O}(\epsilon^2)$

But $A = Z_\varphi - 1$ and so $(B \sim -\alpha \frac{1}{\epsilon} + \mathcal{O}(\alpha^2))$

$$Z_\varphi = 1 + \frac{\alpha}{2} + \mathcal{O}(\alpha^2)$$

With $\epsilon = 6-d$, in $\overline{MS}$

$$Z_\varphi = 1 + \sum_{n=1}^{\infty} \frac{a_n(\alpha)}{\epsilon^n}$$

$$Z_m = 1 + \sum_{n=1}^{\infty} b_n(\alpha)/\epsilon^n$$

$$Z_g = 1 + \sum_{n=1}^{\infty} \frac{c_n(\alpha)}{\epsilon^n}$$

$$\alpha = \frac{g^2}{(4\pi)^3}$$

$$A = Z_\varphi - 1, B = Z_m - 1$$

$$C = Z_g - 1$$

In $\overline{MS}$ we found

$$A = -\frac{1}{6}\alpha + \mathcal{O}(\alpha^2)$$

$$B = -\alpha + \mathcal{O}(\alpha^2)$$

$$C = -\alpha + \mathcal{O}(\alpha^2)$$

and so

$$a_1(x) = -\frac{1}{6}\alpha + \mathcal{O}(\alpha^2)$$

$$b_1(x) = -\alpha + \mathcal{O}(\alpha^2)$$

$$c_1(x) = -\alpha + \mathcal{O}(\alpha^2)$$

with a_n, b_n, c_n all at least $\mathcal{O}(\alpha^2)$ for $n \geq 2$

Claim: Bare fields (φ_0, g_0, m_0 , etc..) must be indep. of μ .

Why?: Physical quantities are independent of μ and so the "original" Lagrangian must be too

* We will use this to compute the Beta fⁿ and an anomalous dim for the theory.

Beta: $d\alpha/d\ln(\mu)$

a. dim: $dm/d\ln(\mu)$

Start with g_0 :

Define

as well as

$$G(\alpha, \epsilon) \equiv \ln(Z_g^2 Z_\varphi^{-3})$$

for which

$$\alpha_0 \equiv \frac{g_0^2}{(4\pi)^3} = Z_g^2 Z_\varphi^{-3} \mu^\epsilon \alpha$$

$$\begin{aligned}
 & \ln[(1+c_n)^2 (1+a_n)^{-3}] \\
 &= 2 \ln(1+c_n) - 3 \ln(1+a_n) \\
 &\approx 2(c_n) - 3a_n \\
 &\approx \frac{1}{\epsilon} (2c_1 - 3a_1) + \mathcal{O}(\alpha^2)
 \end{aligned}$$

Thus,

$$G(\alpha, \epsilon) = \sum_{n=1}^{\infty} \frac{G_n(\alpha)}{\epsilon^n}$$

with

$$\begin{aligned}
 G_1(\alpha) &= -2\alpha + 3\frac{1}{6}\alpha = \alpha\left(\frac{1}{2} - 2\right) + \mathcal{O}(\alpha^2) \\
 G_2(\alpha) &= -\frac{3}{2}\alpha + \mathcal{O}(\alpha^2)
 \end{aligned}$$

Thus, $\ln(\alpha_0) = G(\alpha, \epsilon) + \epsilon \ln(\tilde{\mu}) + \ln(\alpha)$

If α_0 must be independent of μ , then

$$\frac{d \ln(\alpha_0)}{d \ln(\mu)} = 0 = \frac{d G(\alpha, \epsilon)}{d \ln(\mu)} + \epsilon \frac{d \ln(\tilde{\mu})}{d \ln(\mu)} + \frac{d \ln(\alpha)}{d \ln(\mu)}$$

Since

$\mu = \sqrt{4\pi\sigma} e^{-\gamma} \tilde{\mu}$ then

$$\frac{d \ln(\tilde{\mu})}{d \ln(\mu)} = \frac{d}{d \ln(\mu)} \left[\ln\left(\frac{e^{\gamma}}{\sqrt{4\pi\sigma}}\right) + \ln(\mu) \right] = 1$$

$$0 = \frac{d\alpha}{d \ln(\mu)} \frac{\partial G(\alpha, \epsilon)}{\partial \alpha} + \epsilon + \frac{1}{\alpha} \frac{d\alpha}{d \ln(\mu)}$$

$$\frac{\partial}{\partial \alpha} \left(\sum_{n=1}^{\infty} \frac{G_n(\alpha)}{\epsilon^n} \right) = \sum_{n=1}^{\infty} \frac{G'_n(\alpha)}{\epsilon^n}$$

Thus,

$$\alpha \epsilon + \frac{d\alpha}{d \ln(\mu)} \left(\alpha \sum_{n=1}^{\infty} \frac{G'_n(\alpha)}{\epsilon^n} + 1 \right) = 0$$

or,

$$\frac{d\alpha}{d \ln(\mu)} \left(1 + \frac{\alpha G'_1(\alpha)}{\epsilon} + \frac{\alpha G'_2(\alpha)}{\epsilon^2} + \dots \right) + \alpha \epsilon = 0$$

* Physical reasoning: $d\alpha/d \ln(\mu)$ is the rate at which α must vary to compensate for a small change in $\ln(\mu)$.

μ is introduced so that g_0 is dimensionless and

$$\frac{d\alpha}{d\ln(\mu)} = A(\alpha)\epsilon + B(\alpha)$$

$$(\underbrace{A(\alpha)\epsilon + B(\alpha)} + \underbrace{\alpha G'_1(\alpha)A(\alpha) + \dots}_{\text{higher order}}) + \underbrace{\alpha\epsilon}_{\text{higher order}} \stackrel{!}{=} 0$$

$$A(\alpha) = -\alpha$$

$$B(\alpha) = \alpha^2 G'_1(\alpha)$$

$$\frac{d\alpha}{d\ln(\mu)} = -\alpha\epsilon + B(\alpha)$$

$$B(\alpha) = \alpha^2 G'_1(\alpha) \quad (\text{Beta function})$$

• Terms at higher-order in $1/\epsilon$ must also cancel, this determines all other $G'_n(\alpha)$

$$\leadsto G'_2(\alpha) = \alpha G'_1(\alpha)^2$$

$$\text{With } G_2(\alpha) = -3/2\alpha + \mathcal{O}(\alpha^2)$$

$$B(\alpha) = \alpha^2 \left(-3/2 + \mathcal{O}(\alpha) \right) \\ = -3/2\alpha^2 + \mathcal{O}(\alpha^3)$$

which is in the limit $\epsilon \rightarrow 0$

Let us now look at the anomalous dimension

$$m_0 = Z_\psi^{-1/2} Z_m^{1/2} m$$

$$M(\alpha, \epsilon) \equiv \ln(Z_\psi^{-1/2} Z_m^{1/2})$$

$$= -\frac{1}{2} \ln(Z_\psi) + \frac{1}{2} \ln(Z_m)$$

$$= -\frac{1}{2} \ln\left(1 + \frac{a_n}{\epsilon^n}\right) + \frac{1}{2} \ln\left(1 + \frac{b_n}{\epsilon^n}\right)$$

$$= -\frac{1}{2} \sum \frac{a_n(\alpha)}{\epsilon^n} + \frac{1}{2} \sum \frac{b_n(\alpha)}{\epsilon^n}$$

$$= \frac{1}{\epsilon} \left(\frac{1}{2} \right) (b_1(\alpha) - a_1(\alpha)) + \mathcal{O}\left(\frac{1}{\epsilon^2}, \alpha^2\right)$$

$$= \frac{1}{2\epsilon} \left(-\alpha + \frac{1}{6}\alpha \right) = -\frac{5}{12}\alpha \frac{1}{\epsilon} + \mathcal{O}\left(\frac{1}{\epsilon^2}\right)$$

For

$$M(\alpha, \epsilon) \equiv \sum_{n=1}^{\infty} \frac{M_n(\alpha)}{\epsilon^n}$$

$$M_1(\alpha) = -\frac{5}{12}\alpha + \mathcal{O}(\alpha^2)$$

$$\ln(m_0) = M(\alpha, \epsilon) + \ln(m)$$

$$\frac{d\ln(m_0)}{d\ln(\mu)} = \frac{dM(\alpha, \epsilon)}{d\ln(\mu)} + \frac{d\ln(m)}{d\ln(\mu)} = 0$$

$$\frac{d\alpha}{d\ln(\mu)} \left(\frac{M_1'(\alpha)}{\varepsilon} + \frac{M_2'(\alpha)}{\varepsilon^2} + \dots \right) + \frac{1}{m} \frac{dm}{d\ln(\mu)}$$

(Determined above)

$$(-\alpha\varepsilon + \beta(\alpha)) \left(\frac{M_1'(\alpha)}{\varepsilon} + \frac{M_2'(\alpha)}{\varepsilon^2} + \dots \right) + \frac{1}{m} \frac{dm}{d\ln(\mu)} = 0$$

$$\frac{1}{m} \frac{dm}{d\ln(\mu)} = \left(\alpha M_1'(\alpha) + \mathcal{O}\left(\frac{1}{\varepsilon}\right) \right)$$

As we reasoned for $d\alpha/d\ln(\mu)$, $\frac{dm}{d\ln(\mu)}$ should also be finite for $\varepsilon \rightarrow 0$. Thus $\mathcal{O}\left(\frac{1}{\varepsilon}\right) \rightarrow 0$
(Perhaps via $M_i'(\alpha) \rightarrow 0$ or a telescoping series)

$$\therefore \gamma_m(\alpha) \equiv \frac{1}{m} \frac{dm}{d\ln(\mu)} \quad (\text{Anomalous dimension})$$

$$\text{Here } \gamma_m(\alpha) = \alpha M_1'(\alpha) = \alpha \left(-\frac{5}{12} + \mathcal{O}(\alpha) \right)$$

$$\gamma_m(\alpha) = -\frac{5}{12} \alpha + \mathcal{O}(\alpha^2)$$

* Let us now look at the propagator in $\overline{MS}$ renorm scheme,

$$\tilde{\Delta}(k^2) = i \int d^6x e^{-ikx} \langle \Omega | T \{ \psi(x) \psi(0) \} | \Omega \rangle$$

$$\tilde{\Delta}_0(k^2) = i \int d^6x e^{-ikx} \langle \Omega | T \{ \psi_0(x) \psi_0(0) \} | \Omega \rangle$$

$$\tilde{\Delta}_0(k^2) = Z_\psi \tilde{\Delta}(k^2)$$

$$\ln(\tilde{\Delta}_0) = \ln(Z_\psi) + \ln(\tilde{\Delta})$$

$$\frac{d\ln(\tilde{\Delta}_0)}{d\ln(\mu)} = \frac{d\ln(Z_\psi)}{d\ln(\mu)} + \frac{d\ln(\tilde{\Delta})}{d\ln(\mu)} = 0$$

$$\text{Since } \tilde{\Delta}(k^2) = \tilde{\Delta}(k^2; m, \alpha, \mu)$$

$$\frac{d\tilde{\Delta}}{d\ln(\mu)} = \left(\frac{\partial}{\partial \ln(\mu)} + \frac{\partial \alpha}{\partial \ln(\mu)} \frac{\partial}{\partial \alpha} + \frac{\partial m}{\partial \ln(\mu)} \frac{\partial}{\partial m} \right) \tilde{\Delta}$$

$$\frac{d\ln(Z_\psi)}{d\ln(\mu)} + \frac{1}{\tilde{\Delta}(k^2)} \left(\frac{\partial}{\partial \ln(\mu)} + \frac{\partial \alpha}{\partial \ln(\mu)} \frac{\partial}{\partial \alpha} + \frac{\partial m}{\partial \ln(\mu)} \frac{\partial}{\partial m} \right) \tilde{\Delta}(k^2) = 0$$

$$\frac{d \ln(\mu)}{d \ln(\mu)} \quad \Delta(k^2) \left(\frac{d \ln(\mu)}{d \ln(\mu)} \frac{d \ln(\mu)}{d \alpha} \frac{d \ln(\mu)}{d m} \right)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$\therefore x = \frac{\alpha_1(\alpha)}{\varepsilon} + \frac{\alpha_2(\alpha)}{\varepsilon^2}$$

$$\ln(1+x) = \alpha_1/\varepsilon + (\alpha_2(\alpha) - \frac{\alpha_1^2}{2})/\varepsilon^2 + \dots$$

$$\ln(Z_p) = \frac{\alpha_1(\alpha)}{\varepsilon} + \frac{1}{\varepsilon^2} \left(\alpha_2(\alpha) - \frac{\alpha_1^2(\alpha)}{2} \right) + \mathcal{O}\left(\frac{1}{\varepsilon^3}\right)$$

$$\text{and so } \frac{d \ln(Z_p)}{d \ln(\mu)} = \frac{\partial \ln(Z_p)}{\partial \alpha} \frac{d \alpha}{d \ln(\mu)}$$

$$= \left(\frac{\alpha_1'(\alpha)}{\varepsilon} + \frac{1}{\varepsilon} \left(\alpha_2'(\alpha) - 2 \alpha_1(\alpha) \alpha_1'(\alpha) \right) \right) (-\varepsilon \alpha + \beta(\alpha))$$

$$= -\alpha_1'(\alpha) \alpha + \mathcal{O}\left(\frac{1}{\varepsilon}\right) = \frac{d \ln(Z_p)}{d \ln(\mu)}$$

In the limit, $\varepsilon \rightarrow 0$ $\hat{\Delta}(k^2)$ should vary smoothly

• We define the anomalous dimension of the field

$$\gamma_p(\alpha) \equiv \frac{1}{2} \frac{d \ln(Z_p)}{d \ln(\mu)}$$

$$\text{and so } \gamma_p(\alpha) = \frac{1}{2} \left(-\alpha_1'(\alpha) \alpha + \mathcal{O}(\alpha^2) \right) \\ = \frac{1}{2} \left(\frac{1}{\varepsilon} \alpha + \mathcal{O}(\alpha^2) \right)$$

$$\gamma_p(\alpha) = \frac{1}{12} \alpha + \mathcal{O}(\alpha^2)$$

Going back to $\frac{d \ln(\Delta_0)}{d \ln(\mu)} = 0$:

$$2 \hat{\Delta}(k^2) \gamma_p(\alpha) + \left(\frac{\partial}{\partial \ln(\mu)} + \beta(\alpha) \frac{\partial}{\partial \alpha} + \gamma_m(\alpha) \frac{\partial}{\partial m} \right) \hat{\Delta}(k^2) = 0$$

or

$$\left(\frac{\partial}{\partial \ln(\mu)} + \beta(\alpha) \frac{\partial}{\partial \alpha} + \gamma_m(\alpha) \frac{\partial}{\partial m} + 2 \gamma_p(\alpha) \right) \hat{\Delta}(k^2) = 0$$

in the $\varepsilon \rightarrow 0$ limit

(Callan-Symanzik equation)

* Most interesting in the massless limit and with vanishing beta function:

$$m \rightarrow 0 \quad \text{and} \quad \beta(\alpha) = d\alpha/d\ln(\mu) = 0$$

with $\alpha \neq 0$: i.e. there is still coupling

• Suppose $\exists$ an $\alpha = \alpha_*$ s.t.

$$\beta(\alpha_*) = 0, \quad \alpha_* \neq 0$$

For $\alpha = \alpha_*$ and $m = 0$

$$\left(\frac{\partial}{\partial \ln(\mu)} + 2\gamma_\varphi(\alpha_*) \right) \tilde{\Delta}(k^2) = 0$$

$$2\gamma_\varphi \tilde{\Delta} = -\partial \tilde{\Delta} / \partial \ln(\mu)$$

$$\int 2\gamma_\varphi(\alpha_*) d\ln(\mu) = \int \frac{d\tilde{\Delta}}{\tilde{\Delta}}$$

$$\ln(\tilde{\Delta}) + C = -2\gamma_\varphi \ln(\mu)$$

$$e^C \tilde{\Delta} = \mu^{-2\gamma_\varphi}$$

$$\tilde{\Delta}(k^2) = e^C \mu^{-2\gamma_\varphi}$$

integration constant

....

$$\text{Solution is given by: } \tilde{\Delta}(k^2) = \frac{C(\alpha_*)}{k^2} \left(\frac{\mu^2}{k^2} \right)^{-\gamma_\varphi(\alpha_*)}$$

28.1 | φ^4 theory:

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 - \frac{1}{24} Z_\lambda \lambda \mu^{-\epsilon} \varphi^4$$

where is

$$-d + 2 + 2[\varphi] = 0 \Rightarrow [\varphi] = (d-2) \frac{1}{2}$$

$$-d + 2(d-2) + [\lambda] \stackrel{!}{=} 0$$

$$[\lambda] = 4-d$$

So it is renorm in $d=4$

* Let $4-\epsilon = 4 \Rightarrow \epsilon = 4-d$

(a) Compute $\beta(\alpha)$ to order $\mathcal{O}(\lambda^2)$

(b) $\gamma_m(\alpha)$ of m to $\mathcal{O}(\lambda)$

(c) $\gamma_\varphi(\alpha)$ to $\mathcal{O}(\lambda)$

$$\varphi_0 = \sum_{1/2} \varphi$$

$$m_0 = Z_m^{-1/2} m$$

$$\lambda_0 = Z_\lambda \mu^{0\epsilon} Z_\phi^{-2} \lambda$$

* From previous problems:

$$Z_\phi = 1 + \mathcal{O}(\lambda^2)$$

$$Z_\lambda = 1 + \frac{3\lambda}{16\pi^2} \left(\frac{1}{\epsilon} + \ln\left(\frac{\mu}{m}\right) + \frac{1}{3} \right) + \mathcal{O}(\lambda^2)$$

(c) Beta fⁿ $\beta(\alpha)$ of ϕ^4 :

$$\ln(\lambda_0) = \ln(Z_\lambda Z_\phi^{-2}) + \epsilon \ln(\tilde{\mu}) + \ln(\lambda)$$

$$\ln(Z_\lambda Z_\phi^{-2}) \equiv G(\lambda, \epsilon)$$

Now, in $\overline{MS}$

$$Z_{\lambda, \overline{MS}} = 1 + \frac{3\lambda}{16\pi^2} \left(\frac{1}{\epsilon} \right) + \mathcal{O}(\lambda^2)$$

$$G(\lambda, \epsilon) = \mathcal{O}(\lambda^2) + \left(\frac{3\lambda}{16\pi^2} \right) \frac{1}{\epsilon} + \mathcal{O}(\lambda^2)$$

$$= \left(3\lambda/16\pi^2 \right) \frac{1}{\epsilon} + \mathcal{O}(\lambda^2)$$

$$\text{So, for } G(\lambda, \epsilon) \equiv \sum_{n=2}^{\infty} G_n(\lambda) / \epsilon^n$$

$$G_1(\lambda) = \frac{3\lambda}{16\pi^2}$$

$$0 = \frac{d G(\lambda, \epsilon)}{d \ln(\mu)} + \epsilon + \frac{1}{\lambda} \frac{d \lambda}{d \ln(\mu)}$$

$$\frac{d G}{d \ln(\mu)} = \frac{d \lambda}{d \ln(\mu)} \frac{d G(\lambda, \epsilon)}{d \lambda}$$

and so:

$$\lambda \epsilon + \frac{d \lambda}{d \ln(\mu)} \left(1 + \lambda \sum_{n=2}^{\infty} \frac{G'_n(\lambda)}{\epsilon^n} \right) = 0$$

Only in $\epsilon^{\pm 1}$ order:

$$\lambda \epsilon + \frac{d \lambda}{d \ln(\mu)} \left(1 + \frac{\lambda}{\epsilon} \left(\frac{3}{16\pi^2} \right) \right) = 0$$

$$\begin{aligned}\frac{d\lambda}{d\ln(\mu)} &= -\varepsilon\lambda / \left(1 + \frac{\lambda}{\varepsilon} \left(\frac{3}{16\pi^2}\right)\right) \\ &= -\varepsilon\lambda \left(1 - \frac{\lambda}{\varepsilon} \left(\frac{3}{16\pi^2}\right)\right) \\ &= -\varepsilon\lambda + \lambda^2 \left(\frac{3}{16\pi^2}\right)\end{aligned}$$

$\varepsilon \rightarrow 0$

$$\beta(\lambda) = \lambda^2 \left(\frac{3}{16\pi^2}\right) + \mathcal{O}(\lambda^3)$$

⑥ The anomalous dimension $\gamma_m(a) = \frac{1}{m} \frac{dm}{d\ln\mu}$

$$M_0 = Z_d^{-1/2} Z_m^{1/2} m$$

$$Z_m = 1 + \frac{\lambda}{16\pi^2\varepsilon}$$

$$\begin{aligned}M(\lambda, \varepsilon) &= -\frac{1}{2} \ln(Z_d) + \frac{1}{2} \ln(Z_m) \\ &= \mathcal{O}(\lambda^2) + \frac{1}{2} \left(\frac{1}{16\pi^2}\right) \frac{1}{\varepsilon} + \mathcal{O}(\lambda^2) \\ &= \frac{1}{32} \left(\frac{1}{\pi^2}\right) \frac{1}{\varepsilon} + \mathcal{O}(\lambda^2)\end{aligned}$$

$$\ln(M_0) = M(\lambda, \varepsilon) + \ln(m)$$

$$0 = \frac{d\lambda}{d\ln(\mu)} \frac{\partial M}{\partial \lambda} + \frac{1}{m} \frac{dm}{d\ln(\mu)}$$

$$\begin{aligned}\gamma_m(\lambda) &= -\beta(\lambda) \frac{\partial M}{\partial \lambda} \\ &= -\left(-\varepsilon\lambda + \lambda^2 \left(\frac{3}{16\pi^2}\right)\right) \left(\frac{1}{32\pi^2} \left(\frac{1}{\varepsilon}\right) + \mathcal{O}(\lambda^2)\right)\end{aligned}$$

$$\therefore \gamma_m(\lambda) = + \frac{\lambda^2}{32\pi^2} + \mathcal{O}(\lambda^3)$$

⑦ Since $Z_\phi = \mathcal{O}(\lambda^2)$, $\gamma_\phi(\lambda)$ does not contribute to order λ .

$$\gamma_\phi(\lambda) = \mathcal{O}(\lambda^2)$$

28.3 | Now take the theory w two

real scalars:

$$\mathcal{L} = -\frac{1}{2} Z_\phi (\partial_\mu \phi)^2 - \frac{1}{2} Z_m m^2 \phi^2 + Y \phi \\ - \frac{1}{2} Z_\chi (\partial_\mu \chi)^2 - \frac{1}{2} Z_M M^2 \chi^2 \\ + \frac{1}{6} Z_g g \tilde{m}^{3/2} \phi^3 + \frac{1}{2} Z_h h \tilde{M}^{3/2} \phi \chi^2$$

$d = 6 - \epsilon$, ϕ and χ are real scalars
 Y is adjusted to make $\langle \Omega | \phi(x) | \Omega \rangle = 0$

• Y cancels out the tadpole diagrams

$$\left(\begin{array}{c} \text{tadpole diagram} \\ \text{with external line } x \text{ and internal line } w \end{array} \right) \sim \langle \Omega | \phi \int d^6 w \phi_w^3 | \Omega \rangle$$

Since no tadpole terms exist (χ^3)
 we do not need this term for χ .

* We must now look at all 1-loop contributions to both ϕ and to χ for the given scalar terms.